

What is a work system? (M1.3)

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What Is a Work System?

- To understand work system design, we must first understand:
 - What a **system** is in engineering terms
 - Why **systems thinking** differs from component thinking
- Systems thinking focuses on interaction, not isolated parts.

What Is a System?

- A system is:
 - A bounded, goal-oriented set of interrelated elements
 - That transforms inputs into outputs
 - Through structured interaction
 - While maintaining dynamic internal states
- Five key characteristics:
 1. **Purpose** — Exists to achieve an objective
 2. **Boundary** — Distinguishes system from environment
 3. **Inputs/Outputs** — Transforms resources into results
 4. **Interdependence** — Elements influence one another
 5. **State & Feedback** — Internal conditions evolve over time
- Work systems are **open systems**:
 - Exchange information, materials, and energy with environment

Systems Perspective: Emergence

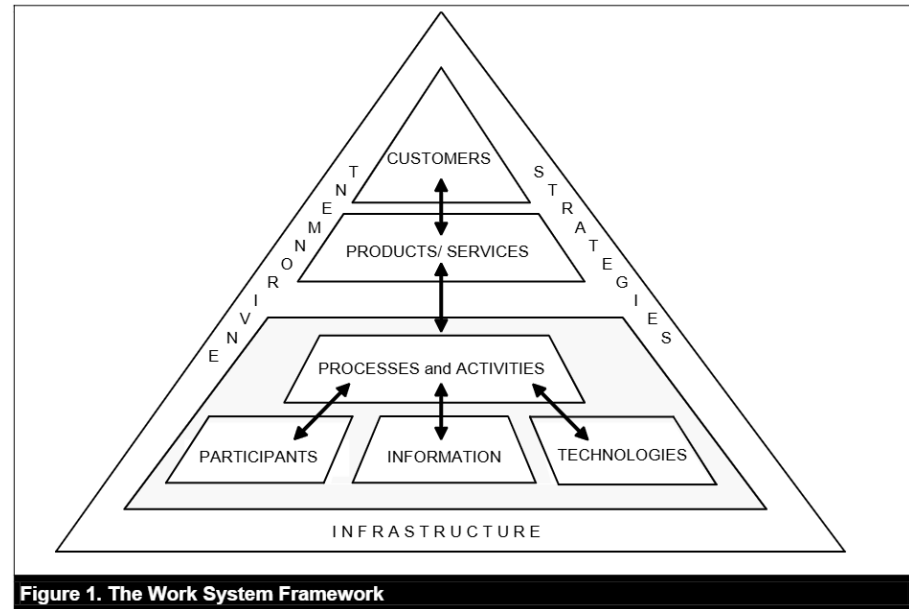
- From a systems perspective:
 - Performance is not located in a single part
 - It **emerges from structured interaction**
- Emergence means:
 - Outcomes arise from how elements work together
 - Not from any single component alone
- This concept is foundational for understanding work systems.

From Systems to Work Systems

- Major Work System Frameworks
 - Alter's Work System Framework
 - SEIPS Model
 - ISO 6385 Ergonomics Framework

Alter's Work System Framework (Managerial Lens)

- Core elements:
 - Customers
 - Products/services
 - Processes and activities
 - Participants
 - Information
 - Technologies
 - Environment
 - Infrastructure
 - Strategy

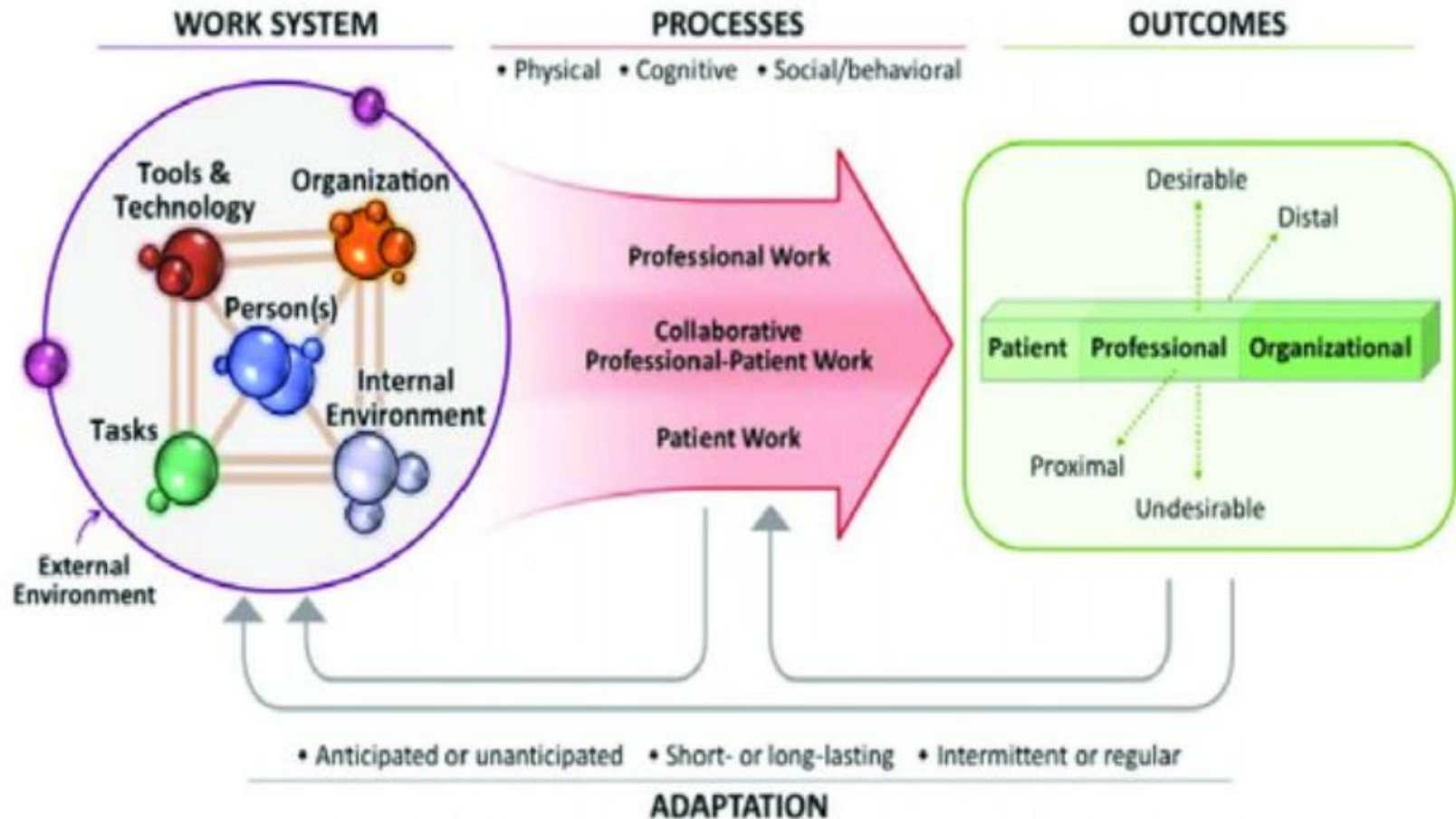


- Core insight:
 - Systems should not be analyzed as just IT
 - Technology is only one component of a broader work system

SEIPS Model (Human Factors Lens)

- SEIPS = Systems Engineering Initiative for Patient Safety
- Core components:
 - Person(s)
 - Tasks
 - Tools/technology
 - Physical environment
 - Organizational conditions
- These influence:
 - Processes
 - Outcomes (safety, quality, well-being)
- Contribution:
 - Explicit link between system design and safety
 - Emphasizes interaction and variability

SEIPS Model (Human Factors Lens)



ISO 6385 (Normative Ergonomic Lens)

- Focus:
 - Balanced work system design considering:
 - Human capabilities
 - Technical requirements
 - Organizational structure
- Contribution:
 - Performance and well-being must be jointly optimized
 - Formalizes socio-technical balance

Convergence and Emergence

- Across frameworks, common components include:
 - Participants
 - Tasks/processes
 - Tools/technology
 - Information
 - Organizational/environmental context
 - Outputs/customers
- Shared assumption:
 - System performance is emergent
- Examples:
 - Productivity → task coordination
 - Safety → demand-capability alignment
 - Quality → stable interactions
 - Motivation → task structure & feedback
- Improvement requires systemic redesign.

Integrated Definition of Work System

- A work system is an open socio-technical system
- Humans and/or machines perform interdependent processes
- Using resources to produce value-generating outputs

Industrial Engineering

Industrial Engineering is concerned with the **DESIGN, IMPROVEMENT and INSTALLATION** of **INTEGRATED SYSTEMS** of **PEOPLE, MATERIAL, EQUIPMENT, INFORMATION and ENERGY**.

It draws upon specialized knowledge and skills in the **MATHEMATICAL, PHYSICAL and SOCIAL SCIENCES** together with the principles and methods of engineering analysis and design to **SPECIFY, PREDICT and EVALUATE** the results to be obtained from such systems.

Distinguishing Features of Work Systems

- Work systems are:
 - **Socio-technical** — Human and technical elements interact
 - **Value-oriented** — Outputs matter to stakeholders
 - **Organizationally embedded** — Roles, rules, and authority shape behavior

Scientific Management in System Terms

- Scientific management can be reinterpreted as:
 - An early attempt at work system redesign
- Taylor aimed to:
 - Reduce variability
 - Standardize methods
 - Clarify roles
 - Separate planning and execution
 - Establish measurement and feedback
- System interpretation:
 - Control internal state
 - Stabilize performance through structured interaction
- Scientific management = early work system engineering

Mini-Summary

- A work system is:
 - An open socio-technical system
 - Organized around purposeful activity
 - Designed to create value
- Performance emerges from interaction among:
 - People
 - Tasks
 - Tools
 - Information
 - Organizational context
- Systems thinking shifts focus from isolated components to structured interaction and feedback